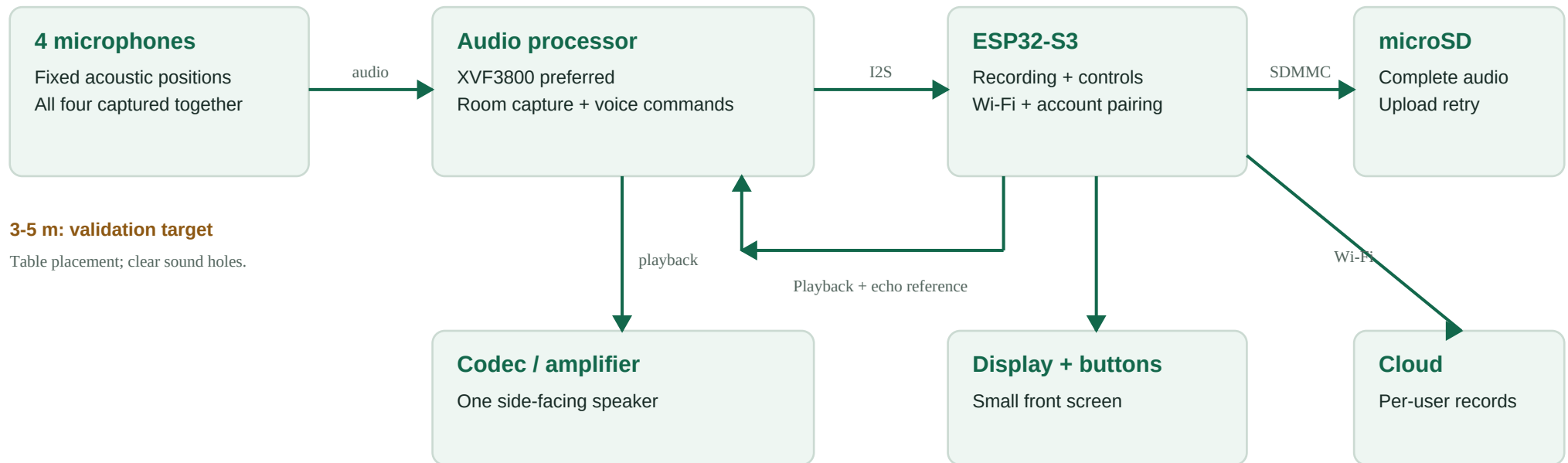


Meeting recorder: custom PCB

Revision A0 | Four microphones, one speaker | 3-5 metre room-pickup target

Engineering draft. This packet defines the circuits and interfaces to design. It is not a completed pin-level schematic, PCB layout or manufacturing file set.



Recommended first build: a custom ESP32/power/storage main board plus a replaceable audio-processing board. Prove room pickup with an existing array before designing four individual microphone heads.

Keep the smaller front display, volume and record buttons, USB-C charging, side-access microSD and a side-facing speaker. Final PCB and case dimensions follow the battery, audio assembly and display selection.

Audio design for 3-5 metre pickup

A microphone hears every audible source. Extra channels enable processing; they do not identify four people automatically.

Route	Audio hardware	Trade-off
A0 - recommended first prototype	ReSpeaker XVF3800 + custom ESP32 main PCB	Existing array and processing. Fastest way to evaluate the 3-5 m target; four microphones are built onto the module.
A1 - custom mechanical design	XVF3800 reference design + four PDM mic heads	MIC1-MIC4 connectors and fixed positions around the case. More PCB design, firmware integration and acoustic tuning.
B - lower-cost research route	ES7210 + four analog mic heads	Useful for synchronized raw capture. Needs more DSP development; not the default room-distance recommendation.

Candidate microphones: IM69D130 PDM heads for A1; IM73A135 differential analog heads for B. These have different interfaces and are not interchangeable. A purchased array uses its own installed microphones.

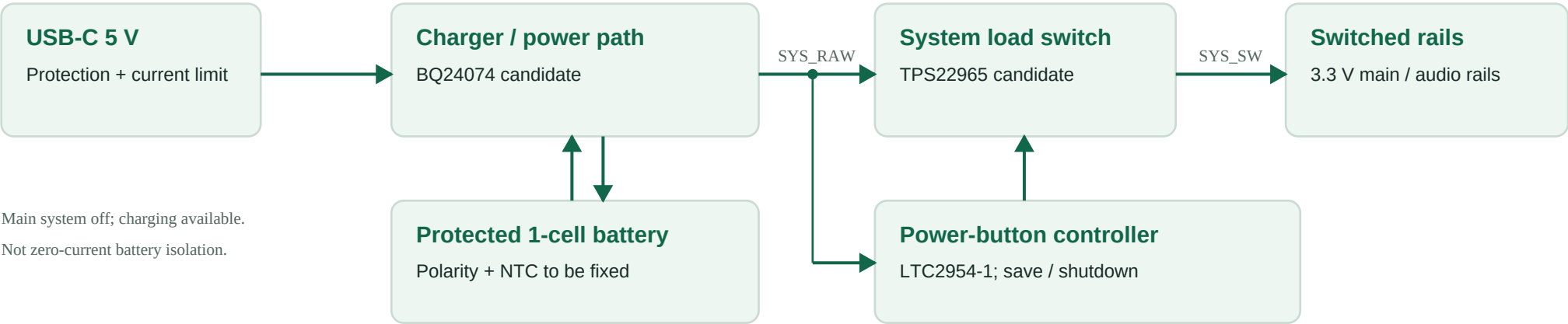
Preserve conversation: evaluate conference and command-processing outputs separately. Strong voice isolation can suppress a quiet participant; test alternating and overlapping speakers. Keep all four raw inputs only where the selected interface supports them.

Speaker echo: send the same playback signal through the speaker path and the processor reference input. The firmware must match clock roles, sample rates and channel format. Existing firmware needs a new audio-board driver.

Basis: [XMOS XVF3800](#) and [Sseed interface documentation](#). Published range is not a guarantee in this enclosure.

A real power button with safe file closure

The processor, SD, display and audio loads switch off. Charging remains available while the device is off.



Circuit	Design requirement
USB-C input	5 V sink, individual 5.1 kohm Rd on CC1 and CC2, ESD protection and verified input-current control. Isolate data/off-state back-power paths.
Battery + charger	Protected one-cell pack; confirm polarity, thermistor, charge limits and thermal performance. BQ24074 is a candidate, not a released circuit.
Main / audio supply	TPS63070 candidate for 3.3 V. A0 audio module needs an additional regulated, switched 5 V stage; its part and load budget are pending.
Power button	LTC2954-1 + TPS22965 candidates: stop capture, finalize/flush SD, persist pending upload, mute, then remove system power. Forced-off recovery also needs testing.

Do not wait for an internet upload to power off. Persistent recovery after a power cut still needs firmware work. A charger and button controller draw some standby current; this arrangement is not a mechanical battery disconnect.

A GPIO budget that leaves the memory pins alone

Proposed ESP32-S3-WROOM-1-N16R8 allocation. These are GPIO numbers, not physical package-pad numbers.

Function	GPIO(s)	Constraint
Audio MCLK reserve	4	Only if required by the selected DSP firmware
Audio BCLK / frame sync	5 / 7	Choose one clock master; finalize firmware image
Audio capture / playback	6 / 8	ESP RX / ESP TX; playback also feeds echo reference
Audio reset; I2C SDA / SCL	9; 16 / 15	Check control addresses, voltage and off-state paths
LCD MOSI / SCLK / CS	11 / 12 / 10	SPI display; final panel/FPC still to select
LCD DC / RESET / backlight	13 / 14 / 47	Backlight uses a driver stage
SD CLK / CMD / D0	38 / 40 / 39	Dedicated SDMMC wiring
SD D1 / D2 / D3	41 / 42 / 48	Four-bit target; route beside ground reference
Volume + / - / Record	1 / 2 / 21	Switches with defined pull-ups and debounce
Power request / acknowledge	17 / 18	Power-controller handshake; verify boot defaults
USB D- / D+	19 / 20	USB programming; ESD and off-state isolation
UART TX / RX; BOOT / RESET	43 / 44; 0 / EN	Recessed service controls and test pads

Keep GPIO35-37 reserved for this module's octal PSRAM. Preserve GPIO0 boot behaviour; avoid peripheral loads on strap pins 3/45/46. A hardware designer must verify module-pad mapping, clocks, power domains and reset behaviour before capture/layout.

Components for an assembled prototype

Candidate parts for quoting. A0 and A1 audio rows are alternatives, not components to fit together.

Block	Candidate / assembly	Before release
Processor	ESP32-S3-WROOM-1-N16R8	Antenna placement, decoupling and module pad mapping
Audio	A0 module OR A1 XVF3800 + four PDM heads	Select reference design, firmware image, connector and acoustic geometry
Power	BQ24074, LTC2954-1, TPS22965, TPS63070	Complete calculations, input policy, passives, thermals and isolation
Audio-module supply	Additional switched 5 V converter for A0	Exact part, inductor and current budget pending
Battery reporting	MAX17048 candidate	Pack characterization and off-state I2C isolation
Screen	1.3-inch SPI / ST7789 direction	Exact panel SKU, connector and backlight driver
Storage / USB	DM3AT microSD; USB4105 family candidates	Exact suffixes, footprints and mechanical clearances
Connectors	Keyed battery, audio, display, speaker; A1 MIC1-MIC4	Harness polarity/current rating; no generic interchangeable sockets
Controls	Power, Record/Stop, Volume +/-, BOOT, RESET	Six physical switches; service controls recessed
Speaker / cell	Reuse only after electrical identification	Speaker 41 x 29 x 10 mm; cell 91 x 59 x 5 mm

Ask the assembler to supply the soldered microphone boards and crimped cables. SPK+ and SPK- are an amplified pair; neither speaker wire is ground. A smaller screen alone cannot shrink the product below the current battery footprint.

Local prototype route and release checklist

Research checked 22 September 2026. No order or message has been sent to a supplier.

Lead	Why it is relevant	What to confirm
Cytron Malaysia	ReSpeaker XVF3800 with enclosure, MPN 114993701: RM299; 3 shown when checked.	Live stock, exact revision and access to the integration connector.
Jaavin / Rawang	Advertises PCB design, component sourcing, assembly and prototype testing.	Audio/XMOS experience, 2- and 5-unit quotes, design-file ownership.
JAC Engineering / Cheras + Rawang	Advertises schematic/layout, fabrication, assembly and product testing.	Prototype MOQ, scope, lead time and fabrication location.

Release sequence: acoustic proof with array -> exact components and mechanical envelope -> pin-level schematic and electrical review -> four-layer PCB layout and DRC -> 2-5 assembled prototypes -> power, SD and room-audio acceptance tests.

Files the factory needs: Gerber/drill files for bare boards, plus a full assembly BOM, pick-and-place and assembly drawings. Retain editable KiCad source and a STEP fit model. This PDF and a case STL cannot manufacture the electronics.

Acceptance: record at 1, 3 and 5 m from four directions; repeat with air-conditioning, two voices and speaker playback. Check complete WAVs, upload retries, charging temperature, power-off on USB and recovery after forced shutdown.

Open decisions: A0 module versus A1 custom array; exact cell and speaker ratings; display and connectors; audio power budget; enclosure size. Ask whether both fabrication and assembly are physically in Malaysia.

Links: [Cytron module](#) | [Jaavin](#) | [JAC Engineering](#). The companion SCHEMATIC_SPECIFICATION.md contains the engineering sources; REQUEST_FOR_QUOTATION.txt is ready to share.